SQL Exercise 1

1. Create the table SEMP with the following structure:-

EMPNO	CHAR(4)
EMPNAME	CHAR(20)
BASIC	FLOAT
DEPTNO	CHAR(2)
DEPTHEAD	CHAR(4)

```
mysql> create table e1_semp(empno char(4),empname char(20), basic float,deptno char(2), depthead char(4));
Query OK, 0 rows affected (0.09 sec)
```

2. Create the table SDEPT with the following structure:-

DEPTNO	CHAR(2)
DEPTNAME	CHAR(15)

```
mysql> create table e1_sdept(deptno char(2), deptname char(15));
Query OK, 0 rows affected (0.05 sec)
```

3. Insert into the SDEPT table the following values:-

10, Development
20, Training

```
mysql> insert into e1_sdept values('20','Training');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO SEMP VALUES ('0001','SUNIL',6000,'10',NULL);
ERROR 1146 (42S02): Table 'cdac_kharghar.semp' doesn't exist
mysql> select * from e1_sdept;
+--------+-------------+
| deptno | deptname    |
+--------+-------------+
| 10     | Development |
| 20     | Training    |
+--------+-------------+
2 rows in set (0.00 sec)
```

4. Insert into the SEMP table the following values:-

   0001, SUNIL, 6000, 10
   0002, HIREN, 8000, 20
   0003, ALI, 4000, 10, 0001
   0004, GEORGE, 6000, 0002

```
mysql> INSERT INTO e1_semp VALUES ('0001','SUNIL',6000,'10',NULL);
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO e1_semp VALUES ('0002','HIREN',8000,'20',NULL);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO e1_semp VALUES ('0003','ALI',4000,'10','0001');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO e1_semp VALUES ('0004','GEORGE',6000,'20','0002');
Query OK, 1 row affected (0.02 sec)

mysql> select * from e1_semp;
+-------+---------+-------+--------+----------+
| empno | empname | basic | deptno | depthead |
+-------+---------+-------+--------+----------+
| 0001  | SUNIL   |  6000 | 10     | NULL     |
| 0002  | HIREN   |  8000 | 20     | NULL     |
| 0003  | ALI     |  4000 | 10     | 0001     |
| 0004  | GEORGE  |  6000 | 20     | 0002     |
+-------+---------+-------+--------+----------+
4 rows in set (0.00 sec)
```

Create S, P, J, SPJ tables as specified below and insert a few rows in each table:-

SUPPLIER (S#, Sname, Status, City)	-	S
PARTS (P#, Pname, Color, Weight, City)	-	P
PROJECTS (J#, Jname, City)	-	J
SUPPLIER-PARTS-PROJECT (S#, P#, J#, Qty)	-	SPJ

Sample data for S# column:- 'S1', 'S2', 'S3', etc.
Sample data for P# column:- 'P1', 'P2', 'P3', etc.
Sample data for J# column:- 'J1', 'J2', 'J3', etc.
Sample data for Status column:- 10, 20, 30, etc.

Write the SELECT queries to do the following:-

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```
mysql> CREATE TABLE S(
    -> SNO CHAR(3),
    -> SNAME VARCHAR(20),
    -> STATUS INT,
    -> CITY VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO S VALUES ('S1','Smith',20,'London');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO S VALUES ('S2','Jones',10,'Paris');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO S VALUES ('S3','Blake',30,'Athens');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO S VALUES ('S4','Clark',20,'London');
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE P(
    -> PNO CHAR(3),
    -> PNAME VARCHAR(20),
    -> COLOR VARCHAR(10),
    -> WEIGHT INT,
    -> CITY VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO P VALUES ('P1','Bolt','Red',12,'London');
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO P VALUES ('P2','Nut','Green',17,'Paris');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO P VALUES ('P3','Screw','Blue',14,'London');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO P VALUES ('P4','Cam','Red',13,'Athens');
Query OK, 1 row affected (0.01 sec)
```

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```
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO P VALUES ('P4','Cam','Red',13,'Athens');
Query OK, 1 row affected (0.01 sec)

mysql> CREATE TABLE J(
    -> JNO CHAR(3),
    -> JNAME VARCHAR(20),
    -> CITY VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO J VALUES ('J1','Sorter','Paris');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO J VALUES ('J2','Display','Athens');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO J VALUES ('J3','Console','London');
Query OK, 1 row affected (0.02 sec)

mysql> CREATE TABLE SPJ(
    -> SNO CHAR(3),
    -> PNO CHAR(3),
    -> JNO CHAR(3),
    -> QTY INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql>
mysql> INSERT INTO SPJ VALUES ('S1','P1','J1',200);
Query OK, 1 row affected (0.02 sec)

mysql> INSERT INTO SPJ VALUES ('S2','P2','J2',300);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO SPJ VALUES ('S3','P3','J3',150);
Query OK, 1 row affected (0.01 sec)

mysql>
```

```
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mysql> select * from spj;
+------+------+------+------+
| SNO  | PNO  | JNO  | QTY  |
+------+------+------+------+
| S1   | P1   | J1   |  200 |
| S2   | P2   | J2   |  300 |
| S3   | P3   | J3   |  150 |
+------+------+------+------+
3 rows in set (0.00 sec)

mysql> select * from s;
+------+-------+--------+--------+
| SNO  | SNAME | STATUS | CITY   |
+------+-------+--------+--------+
| S1   | Smith |     20 | London |
| S2   | Jones |     10 | Paris  |
| S3   | Blake |     30 | Athens |
| S4   | Clark |     20 | London |
+------+-------+--------+--------+
4 rows in set (0.00 sec)

mysql> select * from p;
+------+-------+-------+--------+--------+
| PNO  | PNAME | COLOR | WEIGHT | CITY   |
+------+-------+-------+--------+--------+
| P1   | Bolt  | Red   |     12 | London |
| P2   | Nut   | Green |     17 | Paris  |
| P3   | Screw | Blue  |     14 | London |
| P4   | Cam   | Red   |     13 | Athens |
+------+-------+-------+--------+--------+
4 rows in set (0.00 sec)

mysql> select * from j;
+------+---------+--------+
| JNO  | JNAME   | CITY   |
+------+---------+--------+
| J1   | Sorter  | Paris  |
| J2   | Display | Athens |
| J3   | Console | London |
+------+---------+--------+
3 rows in set (0.00 sec)
```

5. Display all the data from the S table.

```
mysql> select * from s;
+------+-------+--------+--------+
| SNO  | SNAME | STATUS | CITY   |
+------+-------+--------+--------+
| S1   | Smith |     20 | London |
| S2   | Jones |     10 | Paris  |
| S3   | Blake |     30 | Athens |
| S4   | Clark |     20 | London |
+------+-------+--------+--------+
4 rows in set (0.00 sec)
```

6. Display only the S# and SNAME fields from the S table.

```
mysql> SELECT SNO, SNAME
    -> FROM S;
+------+-------+
| SNO  | SNAME |
+------+-------+
| S1   | Smith |
| S2   | Jones |
| S3   | Blake |
| S4   | Clark |
+------+-------+
4 rows in set (0.00 sec)
```

7. Display the PNAME and COLOR from the P table for the CITY=”London”.

```
mysql> SELECT PNAME, COLOR
    -> FROM P
    -> WHERE CITY='London';
+-------+-------+
| PNAME | COLOR |
+-------+-------+
| Bolt  | Red   |
| Screw | Blue  |
+-------+-------+
2 rows in set (0.00 sec)
```

8. Display all the Suppliers from London. 9. Display all the Suppliers from Paris or Athens.


```
mysql> SELECT PNAME, COLOR
    -> FROM P
    -> WHERE CITY='London';
+-------+-------+
| PNAME | COLOR |
+-------+-------+
| Bolt  | Red   |
| Screw | Blue  |
+-------+-------+
2 rows in set (0.00 sec)
```


9. Display all the Suppliers from Paris or Athens.

```
mysql> SELECT *
    -> FROM S
    -> WHERE CITY='Paris' OR CITY='Athens';
+------+-------+--------+--------+
| SNO  | SNAME | STATUS | CITY   |
+------+-------+--------+--------+
| S2   | Jones |     10 | Paris  |
| S3   | Blake |     30 | Athens |
+------+-------+--------+--------+
2 rows in set (0.00 sec)
```

10. Display all the Projects in Athens.

```
mysql> SELECT *
    -> FROM J
    -> WHERE CITY='Athens';
+------+---------+--------+
| JNO  | JNAME   | CITY   |
+------+---------+--------+
| J2   | Display | Athens |
+------+---------+--------+
1 row in set (0.00 sec)
```

11. Display all the Partnames with the weight between 12 and 14 (inclusive of both).

```
mysql> SELECT PNAME
    -> FROM P
    -> WHERE WEIGHT BETWEEN 12 AND 14;
+-------+
| PNAME |
+-------+
| Bolt  |
| Screw |
| Cam   |
+-------+
3 rows in set (0.00 sec)
```

12. Display all the Suppliers with a Status greater than or equal to 20.

```
mysql> SELECT *
    -> FROM S
    -> WHERE STATUS >= 20;
+------+-------+--------+--------+
| SNO  | SNAME | STATUS | CITY   |
+------+-------+--------+--------+
| S1   | Smith |     20 | London |
| S3   | Blake |     30 | Athens |
| S4   | Clark |     20 | London |
+------+-------+--------+--------+
3 rows in set (0.01 sec)
```

13. Display all the Suppliers except the Suppliers from London.

```
mysql> SELECT *
    -> FROM S
    -> WHERE CITY <> 'London';
+------+-------+--------+--------+
| SNO  | SNAME | STATUS | CITY   |
+------+-------+--------+--------+
| S2   | Jones |     10 | Paris  |
| S3   | Blake |     30 | Athens |
+------+-------+--------+--------+
2 rows in set (0.00 sec)
```

14. Display only the Cities from where the Suppliers come from.

```
mysql> SELECT DISTINCT CITY
    -> FROM S;
+--------+
| CITY   |
+--------+
| London |
| Paris  |
| Athens |
+--------+
3 rows in set (0.01 sec)
```

15. Assuming that the Part Weight is in GRAMS, display the same in MILLIGRAMS and KILOGRAMS.

```
mysql> SELECT
    -> PNAME,
    -> WEIGHT AS GRAMS,
    -> WEIGHT*1000 AS MILLIGRAMS,
    -> WEIGHT/1000 AS KILOGRAMS
    -> FROM P;
+-------+-------+------------+-----------+
| PNAME | GRAMS | MILLIGRAMS | KILOGRAMS |
+-------+-------+------------+-----------+
| Bolt  |    12 |      12000 |    0.0120 |
| Nut   |    17 |      17000 |    0.0170 |
| Screw |    14 |      14000 |    0.0140 |
| Cam   |    13 |      13000 |    0.0130 |
+-------+-------+------------+-----------+
4 rows in set (0.01 sec)
```